
STATE: COSMIC EPIDEMICS — IS IT REAL?
ENGAGE AVATAR LINK?
AVATAR PROFILE:
DR. INFECTIONISTA
CONFIRM
SIMULATION
STATE

WHY SALMONELLA GETS MEANER IN SPACE

The Trigger? →

ON EARTH · gravity



IN SPACE · microgravity



CALM STILL LIQUID → **What is on?** → VIRULENCE GENES ON → 3× MORE DEADLY (in mice)

mice were infected on Earth with space-grown bacteria

WHY SALMONELLA GETS MEANER IN SPACE

the trigger is the ENVIRONMENT, not space itself

ON EARTH · gravity



HIGH FLUID SHEAR

IN SPACE · microgravity



LOW FLUID SHEAR

CALM STILL LIQUID → Hfq → VIRULENCE GENES ON → 3× MORE DEADLY (in mice)

IT'S THE ENVIRONMENT, NOT SPACE MAGIC

① copy the calm liquid on Earth → [redacted] ② add phosphate ions → [redacted]

mice were infected on Earth with space-grown bacteria

HOW DO WE TEST IF SPACE GERMS ARE MEANER?

STEP 1



GROWN IN SPACE

GROWN ON GROUND

BROUGHT BACK TO EARTH

STEP 2

ON EARTH



1 → 10 → 100 → 1000 germs

STEP 3



SPACE GERMS =
fewer germs needed to make half the group sick
= MORE DEADLY

LD50 = the dose that makes half the group sick.
LOWER = scarier.

THE MICE NEVER WENT TO SPACE — ONLY THE BACTERIA DID.

STOWAWAY MICROBES: WHO'S EATING THE SPACE STATION?

MIR STATION (1986–2001)

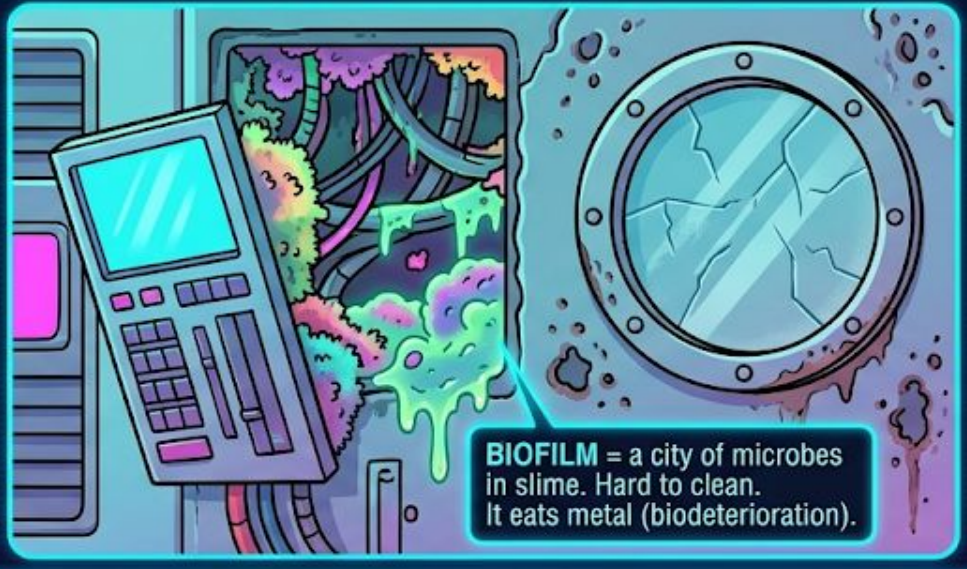


Penicillium Aspergillus



Bacillus

ate the window seal, cables & a comms unit



ISS — water system



Ralstonia pickettii



Burkholderia



Cupriavidus metallidurans

worst in the water-recycling unit

Is it life in space?

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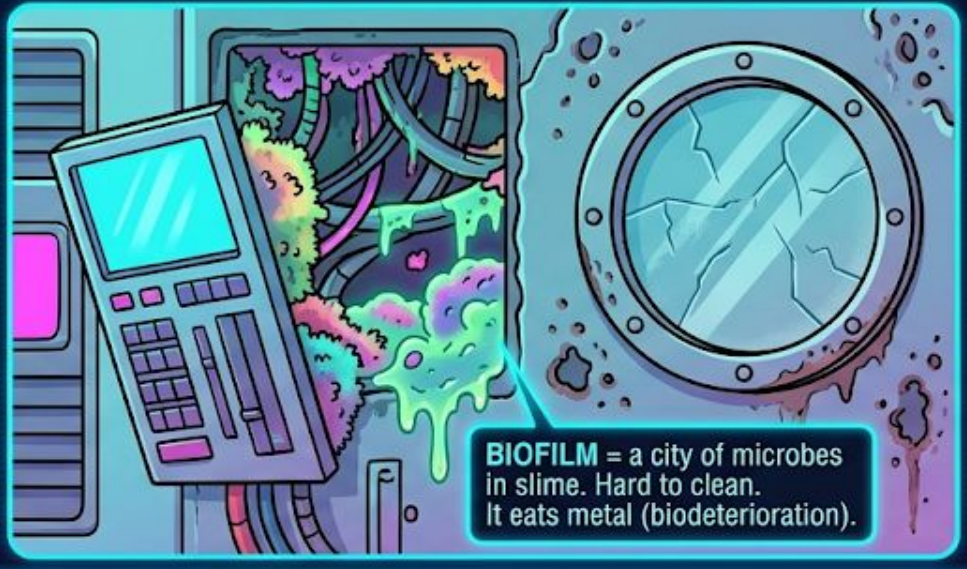


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INSIDE THE STATION THERE'S AIR, WARMTH & WATER —
so microbes thrive, just like on Earth.